

MICROCONTROLLER-BASED MORSE CODE ENCODER



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Morse code is a method of transmitting textual information as a series of on-off tones, lights or clicks that can be directly understood by a trained listener without using any special equipment. The International Morse Code encodes alphabets, numerals and a small set of

punctuation and procedural signals as standardised sequences of short and long signals called 'dots' and 'dashes,' respectively.

Each character (letter or numeral) is represented by a unique sequence of dots and dashes. The duration of a dash is three-times the duration of a dot. Each dot or dash is followed by a short silence, equal to the dot-duration.

The dot-duration is the basic unit of time measurement in code transmission. Morse code speed is measured in words per minute (wpm). Characters have differing lengths as these contain differing numbers of dots and dashes. Consequently, words also have different lengths depending on the total number of dots and dashes, even if these contain the same number of characters.

The Morse code encoder presented here converts texts, numbers and special characters into Morse code as audible output on a piezobuzzer (PZ1). The encoder along with any Morse code transmitter can be used for transmission of Morse code.

Fig. 1 shows the block diagram of the Morse code encoder. The Morse code message is entered through a

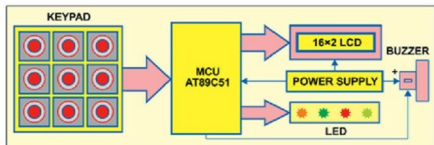


Fig. 1: Block diagram of Morse code encoder

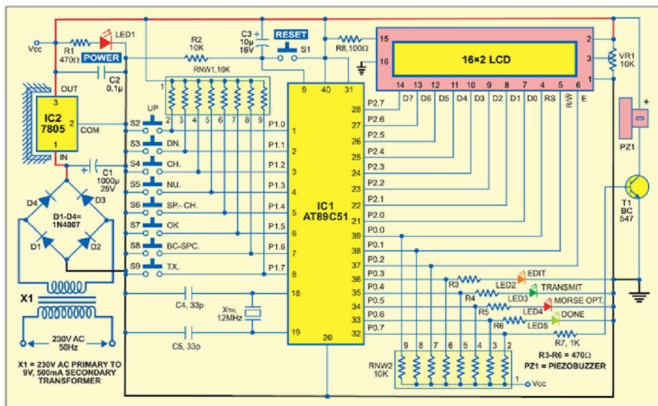


Fig. 2: Circuit of Morse code encoder